

NewStart AI — Government Document Routing (Demo)

Research Results Historical Random Form Demo Family-Aware Robustness Demo

Saved frozen evaluation replay — Version 6 family-aware robustness study (Checkpoint 10). Every prediction below was already computed and saved to disk; this page never calls Gemini, never recomputes an embedding, and never re-runs BERT. Samples only from the 99-document family-disjoint frozen test set, a dataset entirely separate from the historical demo above.

Condition:

Random Family-Aware Document

Routing method for guidance:

Document 542

Effective family: **IRS:W4V** | Ground truth: **IRS** | Condition: **middle_only_masked** (masked, middle)

Source: *saved_frozen_evaluation*

the payer eligible for voluntary withholding. will withhold 10 % from each payment. you aren ' t required to have federal income tax withheld from line 6. if you receive any of the payments listed on line 6, check the these payments. your request is voluntary. box to indicate the percentage (7 %, 10 %, 12 %, or 22 %) you want withheld from each payment. note : payers may develop their own form for you to request federal income tax withholding. if a payer gives you its own form instead of line 7. see how do i stop withholding? below. form w - 4v, use that form. sign this form. form w - 4v is not considered valid unless you sign why should i request withholding? it. you may find that having federal income tax withheld from the listed when will my withholding start? payments is more convenient than making quarterly estimated tax ask your payer exactly when income tax withholding will begin. the payments. however, if you have other income that isn ' t subject to federal income tax withholding you choose on this form will remain withholding, consider making estimated tax payments. for more in effect until you change or stop it or the payments stop. details, see form 1040 - es, estimated tax for individuals. how do i change withholding? how much can i have withheld? if you are getting a payment other than unemployment for unemployment compensation, the payer is permitted to withhold compensation and want to change your withholding rate, complete 10

Saved Checkpoint 10 replay

Method	Predicted agency	Correct	Retrieval similarity	Latency (ms)
bert	IRS	Correct	—	—
llm	IRS	Correct	—	1686
llm_rag	IRS	Correct	0.763	1353

Single-condition live demonstration (Checkpoint 12B)

Run live inference on this document

Single-condition live result

Live demonstration only. These outputs are not included in the frozen Checkpoint 10 evaluation, official metrics, or research artifacts.

Text fingerprint matches registered condition text:
4546279e610a4928c17b95f85489e76d5d117086a92f4a0a62ac536a756e0026

Method	Status	Predicted agency	Correct	BERT probability	Retrieval similarity	Latency (ms)	Completed at
bert	success	IRS	Correct	0.994	—	159	2026-08-04T00:50:51.240588+00:00

Method	Status	Predicted agency	Correct	BERT probability	Retrieval similarity	Latency (ms)	Completed at
llm	success	IRS	Correct	—	—	1813	2026-08-04T00:50:53.056782+00:00
llm_rag	success	IRS	Correct	—	0.763	1401	2026-08-04T00:50:54.518961+00:00

All ten conditions for this document

Saved frozen evaluation replay — no live inference and no API cost

10-condition summary

Method	Correct conditions	Incorrect	Failed / invalid
bert	9/10	1	0
llm	10/10	0	0
llm_rag	10/10	0	0

Per-condition comparison

complete_unmasked

unmasked

Fingerprint: ab5ffbae471aedca...

All methods use the same registered source condition. BERT processes the complete text using its frozen chunking and document-level aggregation policy, while both Gemini methods receive the same main query text under the frozen 6,000-character limit. LLM+RAG additionally receives approved retrieved context.

Method	Status	Predicted	Correct	BERT prob.	Retrieval sim.	Latency (ms)	Routed guidance
bert	success	IRS	Correct	—	—	—	
llm	success	IRS	Correct	—	—	1937	This document appears to relate to the IRS. Review the complete document and the IRS's official instructions or website before taking action.
llm_rag	success	IRS	Correct	—	0.768	1725	

beginning_only_unmasked unmasked

Fingerprint: 1d69ba484293510a...

All three methods begin with the same registered excerpt, subject to their own frozen tokenization and prompt structures. LLM+RAG additionally receives approved retrieved context.

Method	Status	Predicted	Correct	BERT prob.	Retrieval sim.	Latency (ms)	Routed guidance
bert	success	SSA	Incorrect	—	—	—	
llm	success	IRS	Correct	—	—	1528	This document appears to relate to the IRS. Review the complete document and the IRS's official instructions or website before taking action.
llm_rag	success	IRS	Correct	—	0.763	1908	

middle_only_unmasked unmasked

Fingerprint: 6a88a90340b0b998...

All three methods begin with the same registered excerpt, subject to their own frozen tokenization and prompt structures. LLM+RAG additionally receives approved retrieved context.

Method	Status	Predicted	Correct	BERT prob.	Retrieval sim.	Latency (ms)	Routed guidance
bert	success	IRS	Correct	—	—	—	
llm	success	IRS	Correct	—	—	1729	This document appears to relate to the IRS. Review the complete document and the IRS's official instructions or website before taking action.
llm_rag	success	IRS	Correct	—	0.761	1820	

end_only_unmasked unmasked

Fingerprint: 7742bfe9a32e96c5...

All three methods begin with the same registered excerpt, subject to their own frozen tokenization and prompt structures. LLM+RAG additionally receives approved retrieved context.

Method	Status	Predicted	Correct	BERT prob.	Retrieval sim.	Latency (ms)	Routed guidance
bert	success	IRS	Correct	—	—	—	

Method	Status	Predicted	Correct	BERT prob.	Retrieval sim.	Latency (ms)	Routed guidance
llm	success	IRS	Correct	—	—	1928	This document appears to relate to the IRS. Review the complete document and the IRS's official instructions or website before taking action.
llm_rag	success	IRS	Correct	—	0.764	4721	

beginning_middle_end_unmasked unmasked

Fingerprint: e52c834ca8b3028c...

All three methods begin with the same registered excerpt, subject to their own frozen tokenization and prompt structures. LLM+RAG additionally receives approved retrieved context.

Method	Status	Predicted	Correct	BERT prob.	Retrieval sim.	Latency (ms)	Routed guidance
bert	success	IRS	Correct	—	—	—	This document appears to relate to the IRS. Review the complete document and the IRS's official instructions or website before taking action.
llm	success	IRS	Correct	—	—	1579	
llm_rag	success	IRS	Correct	—	0.792	2108	

complete_masked masked

Fingerprint: 7de1aa461ae2fc52...

All methods use the same registered source condition. BERT processes the complete text using its frozen chunking and document-level aggregation policy, while both Gemini methods receive the same main query text under the frozen 6,000-character limit. LLM+RAG additionally receives approved retrieved context.

Method	Status	Predicted	Correct	BERT prob.	Retrieval sim.	Latency (ms)	Routed guidance
bert	success	IRS	Correct	—	—	—	
llm	success	IRS	Correct	—	—	1368	This document appears to relate to the IRS. Review the complete document and the IRS's official instructions or website before taking action.
llm_rag	success	IRS	Correct	—	0.799	986	

beginning_only_masked masked

Fingerprint: 7ae7c2dcba582cd7...

All three methods begin with the same registered excerpt, subject to their own frozen tokenization and prompt structures. LLM+RAG additionally receives approved retrieved context.

Method	Status	Predicted	Correct	BERT prob.	Retrieval sim.	Latency (ms)	Routed guidance
bert	success	IRS	Correct	—	—	—	

Method	Status	Predicted	Correct	BERT prob.	Retrieval sim.	Latency (ms)	Routed guidance
llm	success	IRS	Correct	—	—	1392	This document appears to relate to the IRS. Review the complete document and the IRS's official instructions or website before taking action.
llm_rag	success	IRS	Correct	—	0.778	2363	

middle_only_masked masked

Fingerprint: 4546279e610a4928...

All three methods begin with the same registered excerpt, subject to their own frozen tokenization and prompt structures. LLM+RAG additionally receives approved retrieved context.

Method	Status	Predicted	Correct	BERT prob.	Retrieval sim.	Latency (ms)	Routed guidance
bert	success	IRS	Correct	—	—	—	This document appears to relate to the IRS. Review the complete document and the IRS's official instructions or website before taking action.
llm	success	IRS	Correct	—	—	1686	
llm_rag	success	IRS	Correct	—	0.763	1353	

end_only_masked masked

Fingerprint: 60c5b3814e2e6baa...

All three methods begin with the same registered excerpt, subject to their own frozen tokenization and prompt structures. LLM+RAG additionally receives approved retrieved context.

Method	Status	Predicted	Correct	BERT prob.	Retrieval sim.	Latency (ms)	Routed guidance
bert	success	IRS	Correct	—	—	—	
llm	success	IRS	Correct	—	—	1371	This document appears to relate to the IRS. Review the complete document and the IRS's official instructions or website before taking action.
llm_rag	success	IRS	Correct	—	0.776	1643	

beginning_middle_end_masked masked

Fingerprint: a701e7aa214ec80e...

All three methods begin with the same registered excerpt, subject to their own frozen tokenization and prompt structures. LLM+RAG additionally receives approved retrieved context.

Method	Status	Predicted	Correct	BERT prob.	Retrieval sim.	Latency (ms)	Routed guidance
bert	success	IRS	Correct	—	—	—	

Method	Status	Predicted	Correct	BERT prob.	Retrieval sim.	Latency (ms)	Routed guidance
llm	success	IRS	Correct	—	—	1274	This document appears to relate to the IRS. Review the complete document and the IRS's official instructions or website before taking action.
llm_rag	success	IRS	Correct	—	0.800	1780	

All-ten-conditions live demonstration

Run all 10 conditions live